

BH presentation

06.12.2026

MRI acquisition

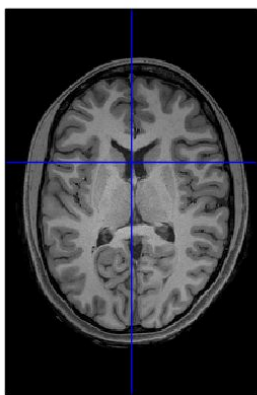
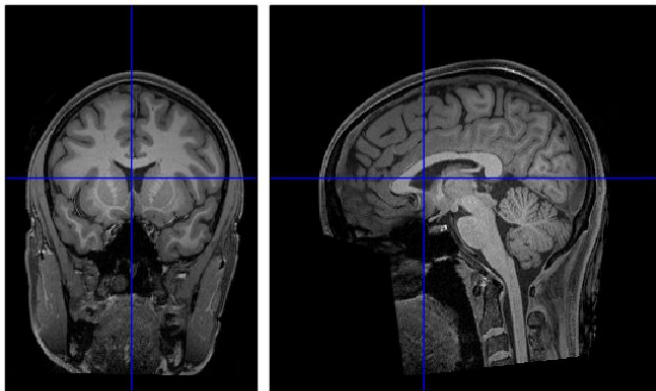
- TR = 2 s
- TE = 14.4, 33.9, 53.4, 72.9, and 92.4 ms
- $2.4 \times 2.4 \times 2.5$ mm resolution
- 48 oblique axial slices with near whole brain coverage
- multiband acceleration 3x
- GRAPPA acceleration 2x
- interleaved slice acquisition)

Which one should be the T1 and T2?

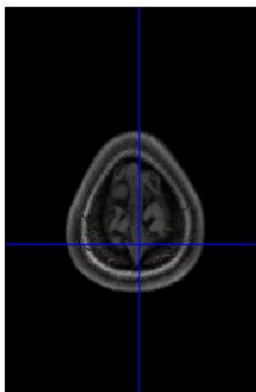
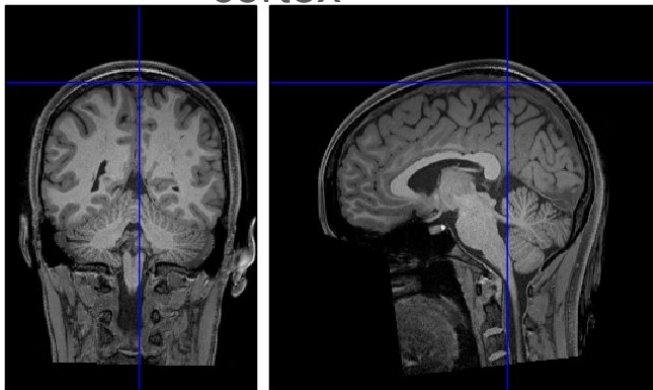
- use check reg to speculate the ventricle, cortex and ghosting artifact.

Check reg: sub1 run1 T1w

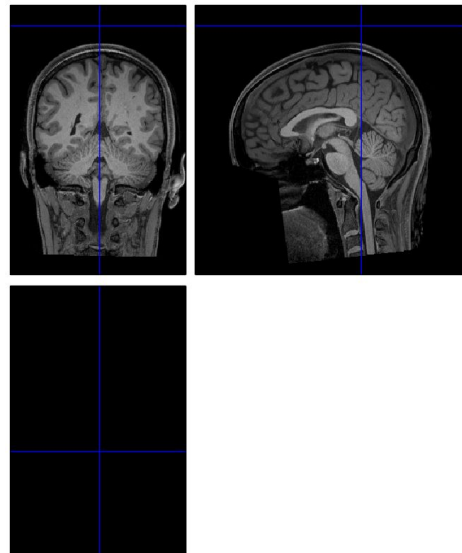
ventricle



cortex

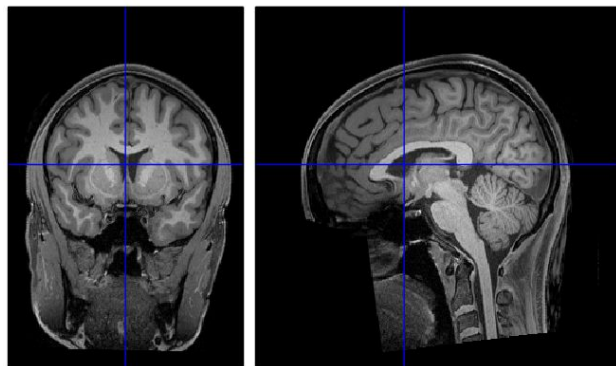


ghosting

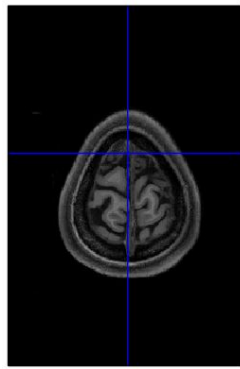
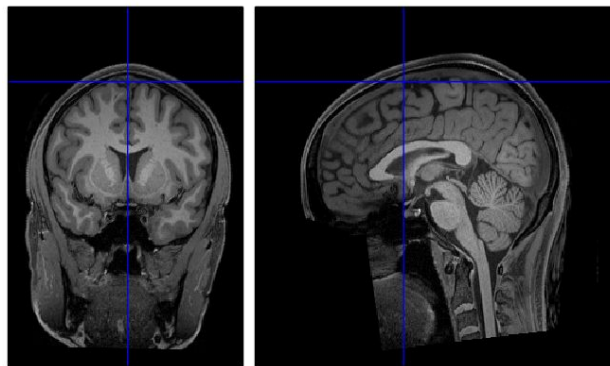


Check reg: sub1 run2 T1w

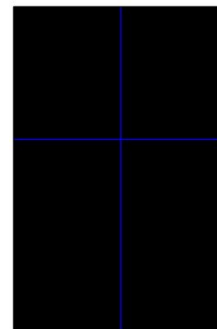
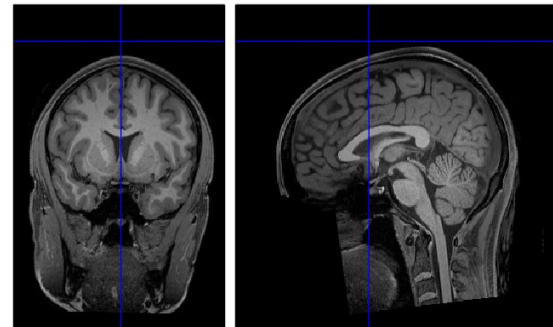
ventricle



cortex

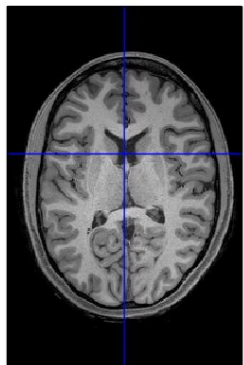
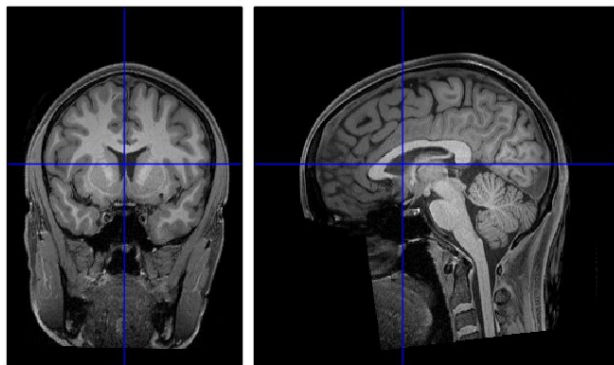


ghosting

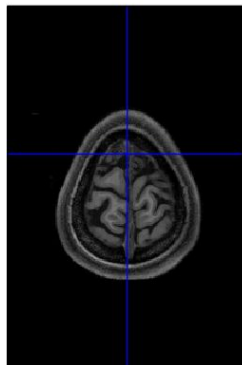
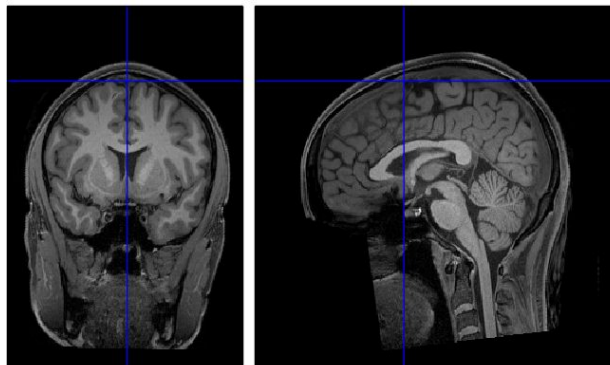


Check reg: sub1 run3 T1w

ventricle



cortex

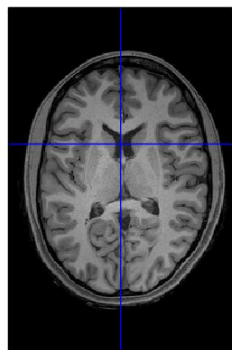
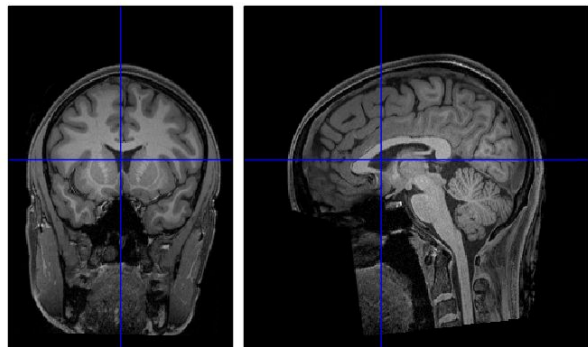


ghosting

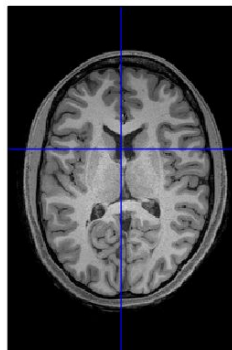
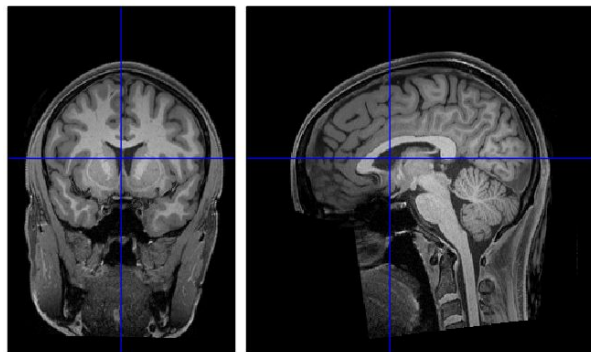


Check reg: sub1 ventricle

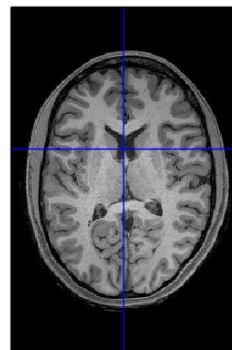
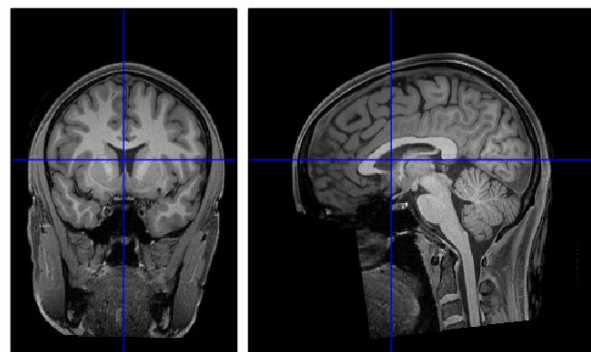
run1



run2

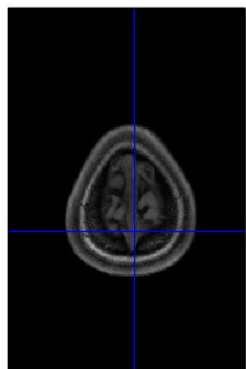
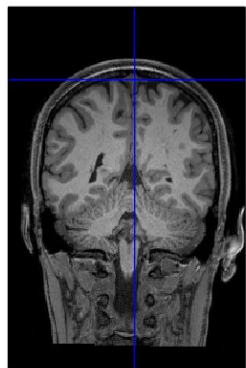


run3

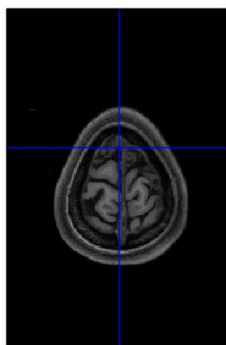
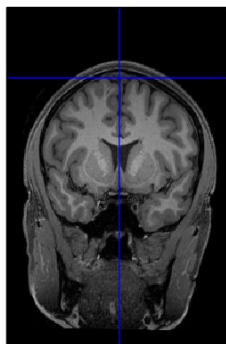


Check reg: sub1 cortex

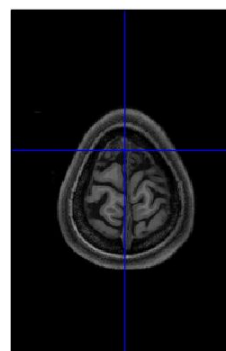
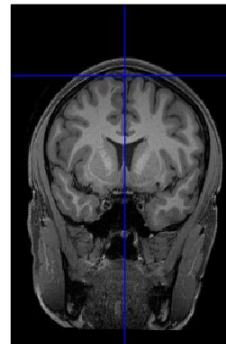
run1



run2

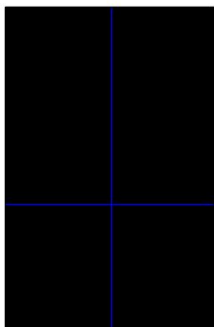
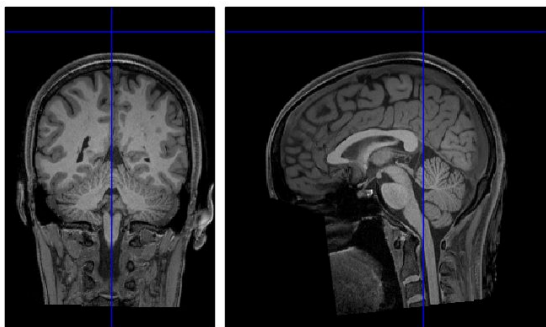


run3

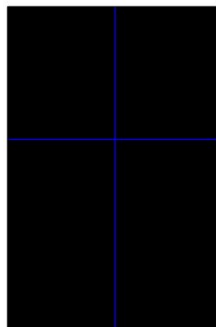
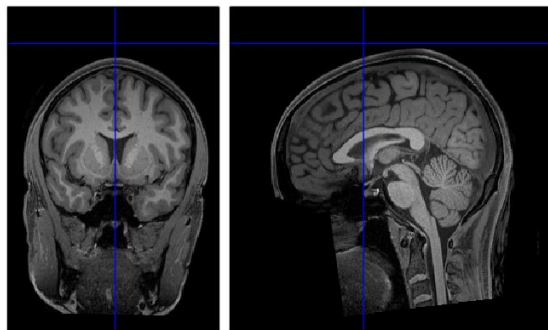


Check reg: sub1 ghosting

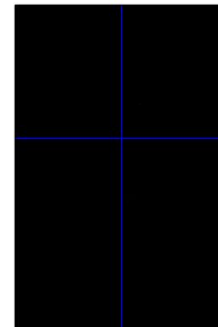
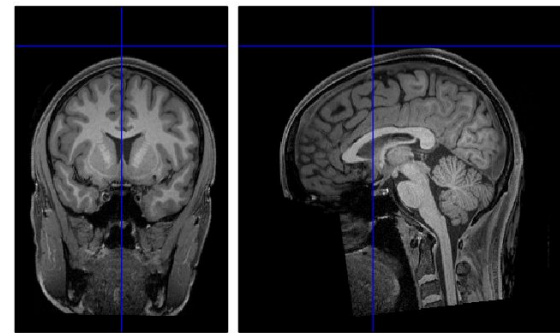
run1



run2

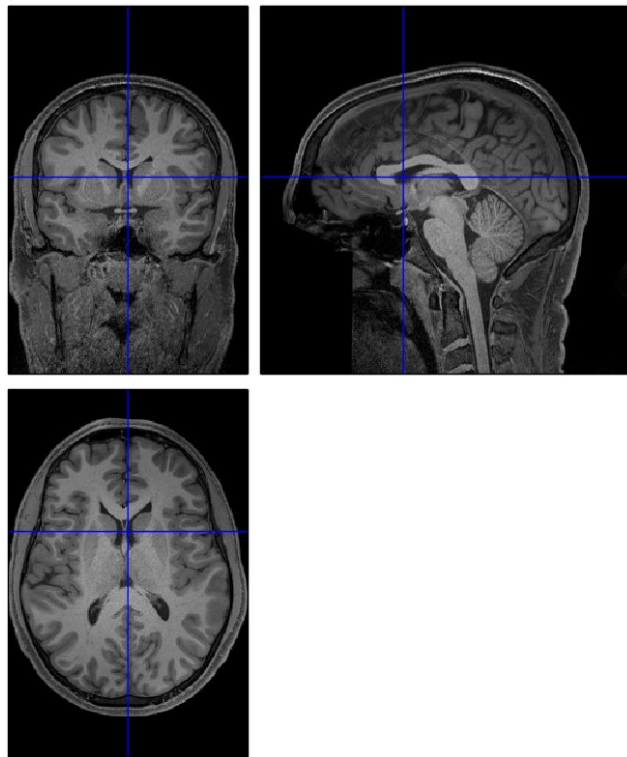


run3



Check reg: sub2 run1 T1

ventricle



cortex

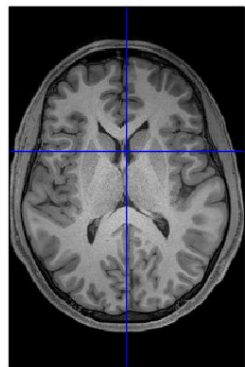
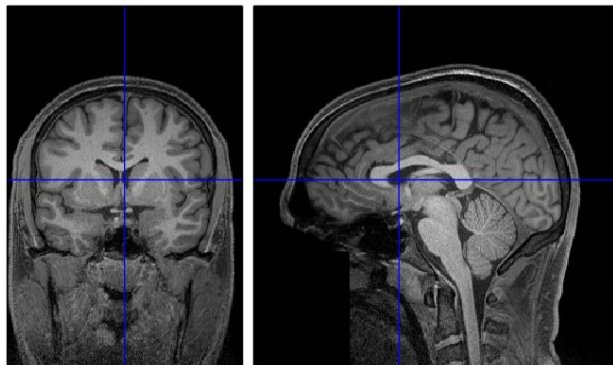


ghosting

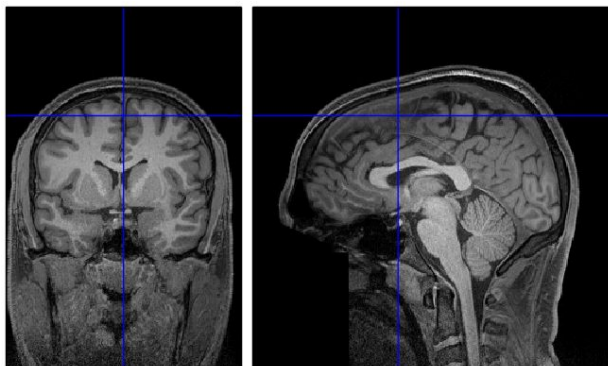


Check reg: sub2 run2 T1

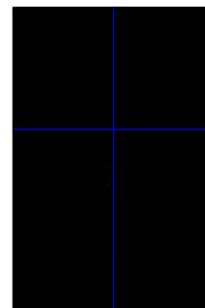
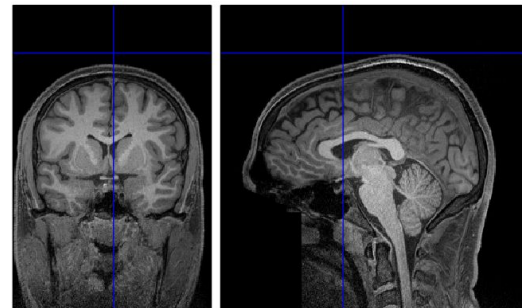
ventricle



cortex

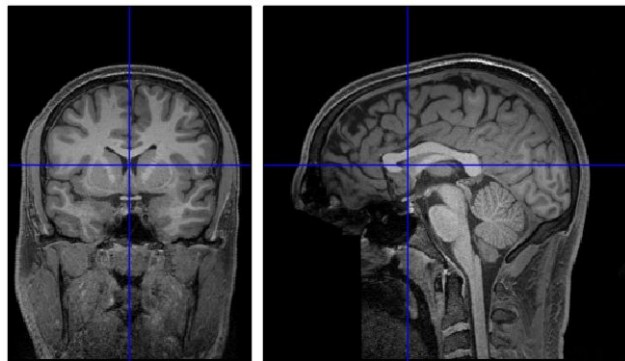


ghosting

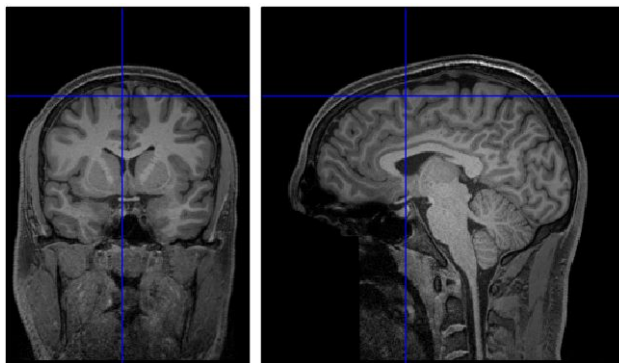


Check reg: sub2 run3 T1

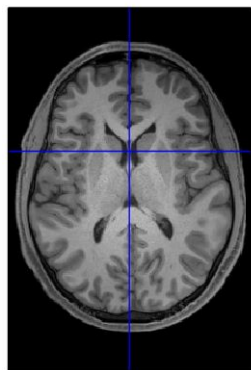
ventricle



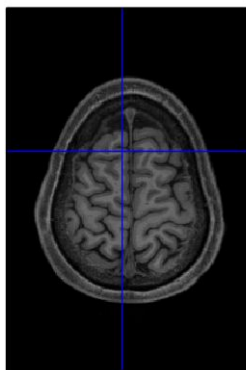
cortex



ghosting



weird ventricle



Strategy: select the run-02. Quality checking:

- sub 1 ok
- sub 2 ok
- sub 3 ok
- sub 4 smaller
- sub 5 bigger
- sub 6 ok
- sub 7 ok
- sub 8 ok
- sub 9 ok
- sub 10 smaller

Preprocessing for 10 subject, too troublesome?

- Create batch for subject 01's preprocessing
- Save batch and script
- Create script for 10 subject's preprocessing

spm script input

外送代號	你的主控制台傳出的東西	參數檔 _job.m 接收後的任務	格式檢查
inputs{1}	epi_runs_paths{1}	送去 Realign 的 Sess 1 (162張時間點)	Cell string (162x1)
inputs{2}	epi_runs_paths{2}	送去 Realign 的 Sess 2 (162張時間點)	Cell string (162x1)
inputs{3}	epi_runs_paths{3}	送去 Realign 的 Sess 3 (162張時間點)	Cell string (162x1)
inputs{4}	epi_runs_paths{4}	送去 Realign 的 Sess 4 (162張時間點)	Cell string (162x1)
inputs{5}	{full_t2_path}	送去 Coregister 的 Source & Ref (高畫質 T2)	1x1 Cell 裝著字串
inputs{6}	{full_t1_path}	送去 Coregister 的 Source 與 Segment	1x1 Cell 裝著字串

Preprocessing

- motion correction
- slice timing
- coregister t2 to epi
- coregister t1 to t2
- segment

- Matlab's SPM in Windows
- Datalad get in Linux

-> expected blocking

-> but not until Dartel create step

Joint Histogram

- Coregistration的derivatives
- 目的: 評估mean EPI 和 T1/T2 之間的對齊程度
- X軸: T1灰階亮度
- Y軸: EPI的灰階亮度
- 圖上每個voxel: 深淺或高度代表多少空間位置同時擁有T1亮度和EPI亮度
- Original joint histogram: 對齊前, mutual information低、entropy (混亂度) 高
- Final joint histogram: 對齊後, SPM透過旋轉、平移將兩張影像結合

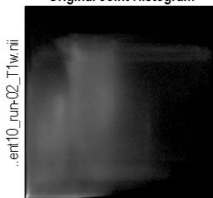
Normalised Mutual Information Coregistration

$$X1 = 0.000^{\circ}X - 1.000^{\circ}Y + 0.002^{\circ}Z + 316.724$$

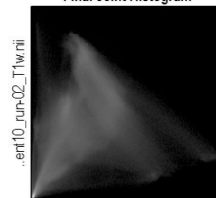
$$Y1 = -0.001^{\circ}X + 0.002^{\circ}Y + 1.000^{\circ}Z + 0.217$$

$$Z1 = 1.000^{\circ}X + 0.000^{\circ}Y + 0.001^{\circ}Z - 0.222$$

Original Joint Histogram



Final Joint Histogram



Derivatives of preprocessing, for each subject

n run: 1-4

n segment: 1-6

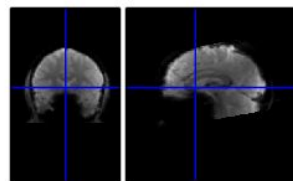
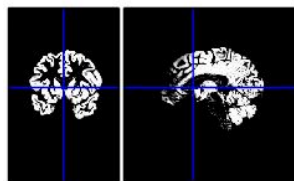
- func/

- rsub-ident01_task-tomloc2_run-01_echo-2_bold.nii: with realign
- arsub-ident01_task-tomloc2_run-01_echo-2_bold.nii: with slice timing after realign
- meansub-ident01_task-tomloc2_run-01_echo-2_bold.nii: 把648 (4x162)跟run-01的第1張epi比較而產出的平均腦
- rp_sub-ident01_task-tomloc2_run-01_echo-2_bold.txt: 此subject在4個run裡面的頭動參數

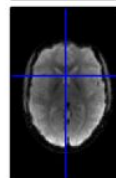
- anat/

- c1sub-ident01_run-02_T1w.nii: segmentation的derivatives, 1-6分別代表 Gray Matter, White Matter, CSF, Bone, Soft Tissue, Air/Background
- rc1sub-ident01_run-02_T1w.nii: r as reslice, 為了跑Dartel的對接影像
- mwc1sub-ident01_run-02_T1w.nii: 標準空間下的實體大腦皮質體積影像 (VBM), w as warped/normalize, m as modulated. 若要比較10個人的voxel-based morphometry才要用
- iy_sub-ident01_run-02_T1w.nii: inverse deformation field, 把MNI返回subject的橋梁

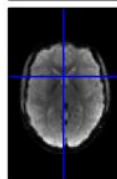
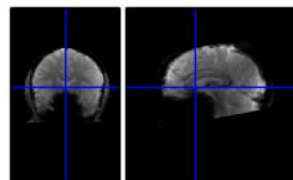
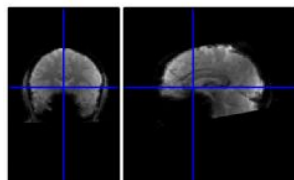
Check reg: sub01



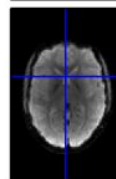
c1



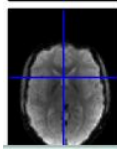
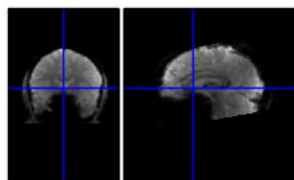
mean



realign

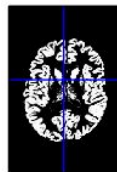
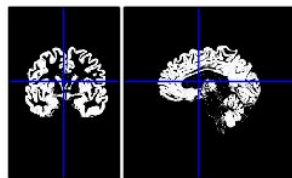


realign+slice timing

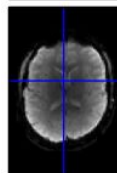
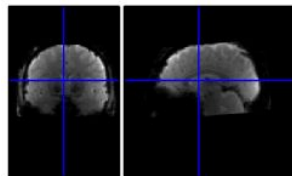


raw

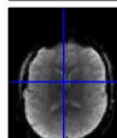
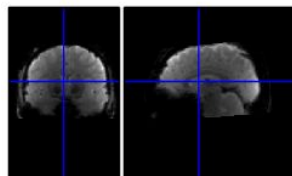
Check reg: sub05



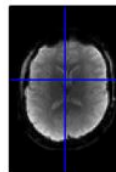
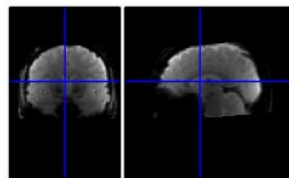
c1



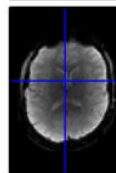
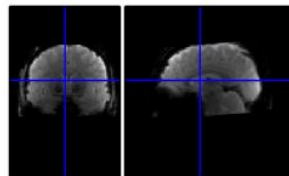
realign



raw

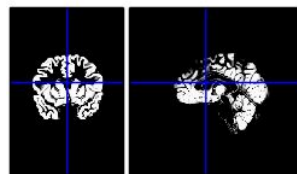


mean

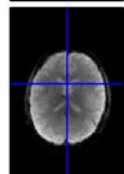
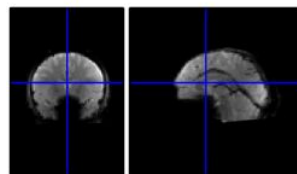


realign+slice timing

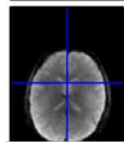
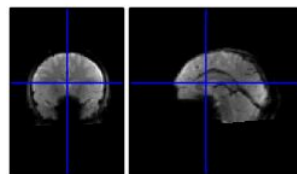
Check reg: sub10



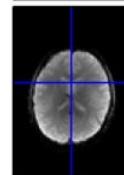
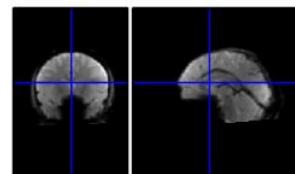
c1



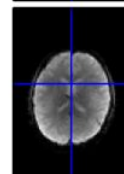
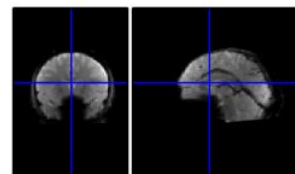
realign



raw



mean



realign+slice timing

Dartel create SST: failed

SPM in Windows

lack of perssion to write file in Linux

05-Jun-2026 16:33:12 - Running 'Run Dartel (create Templates)'

```
Continuing registration from pre-existing parameters (\\wsl$\\Ubuntu\\home\\chsinchi\\subjects_nii\\ds003814\\sub-ident01\\anat\\u_rc1sub-ident01_run-
Continuing registration from pre-existing parameters (\\wsl$\\Ubuntu\\home\\chsinchi\\subjects_nii\\ds003814\\sub-ident02\\anat\\u_rc1sub-ident02_run-
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Continuing registration from pre-existing parameters (\\wsl$\\Ubuntu\\home\\chsinchi\\subjects_nii\\ds003814\\sub-ident07\\anat\\u_rc1sub-ident07_run-
Continuing registration from pre-existing parameters (\\wsl$\\Ubuntu\\home\\chsinchi\\subjects_nii\\ds003814\\sub-ident08\\anat\\u_rc1sub-ident08_run-
Continuing registration from pre-existing parameters (\\wsl$\\Ubuntu\\home\\chsinchi\\subjects_nii\\ds003814\\sub-ident09\\anat\\u_rc1sub-ident09_run-
Continuing registration from pre-existing parameters (\\wsl$\\Ubuntu\\home\\chsinchi\\subjects_nii\\ds003814\\sub-ident10\\anat\\u_rc1sub-ident10_run-
7.82979, 0.259089, 0.0410591, 0.00790892, 0.00303277, 0.00157642, 0.000967744, 0.000658105,
1 1      710553 0      710553 236.651
```

05-Jun-2026 16:34:56 - Failed 'Run Dartel (create Templates)'

Error using [mat2file](#)

Can't open file for writing:

\\wsl\$\\Ubuntu\\home\\chsinchi\\subjects_nii\\ds003814\\sub-ident01\\anat\\u_rc1sub-ident01_run-02_T1w_Template.nii

Check for write permission or whether the directory exists.

In file "C:\\Users\\cindy\\Downloads\\spm\\@file_array\\subsasgn.m", function "subfun" at [line 162](#).

In file "C:\\Users\\cindy\\Downloads\\spm\\@file_array\\subsasgn.m", function "subsasgn" at [line 83](#).

In file "C:\\Users\\cindy\\Downloads\\spm\\@nifti\\subsasgn.m", function "fun" at [line 72](#).

In file "C:\\Users\\cindy\\Downloads\\spm\\@nifti\\subsasgn.m", function "subsasgn" at [line 18](#).

In file "C:\\Users\\cindy\\Downloads\\spm\\@nifti\\subsasgn.m", function "subsasgn" at [line 35](#).

In file "C:\\Users\\cindy\\Downloads\\spm\\toolbox\\DARTEL\\spm_dartel_template.m", function "spm_dartel_template" at [line 172](#).

The following modules did not run:

Failed: Run Dartel (create Templates)

Fixed

- copy all the subjects' rc1~rc10 nifti files to Windows desktop

Dartel normalize MNI

- Voxel size: 2.4 2.4 2.5
- Gaussian FWHM [7 7 7] ($2.4 \cdot (8/3) \sim 6.4$)

Dartel normalize MNI

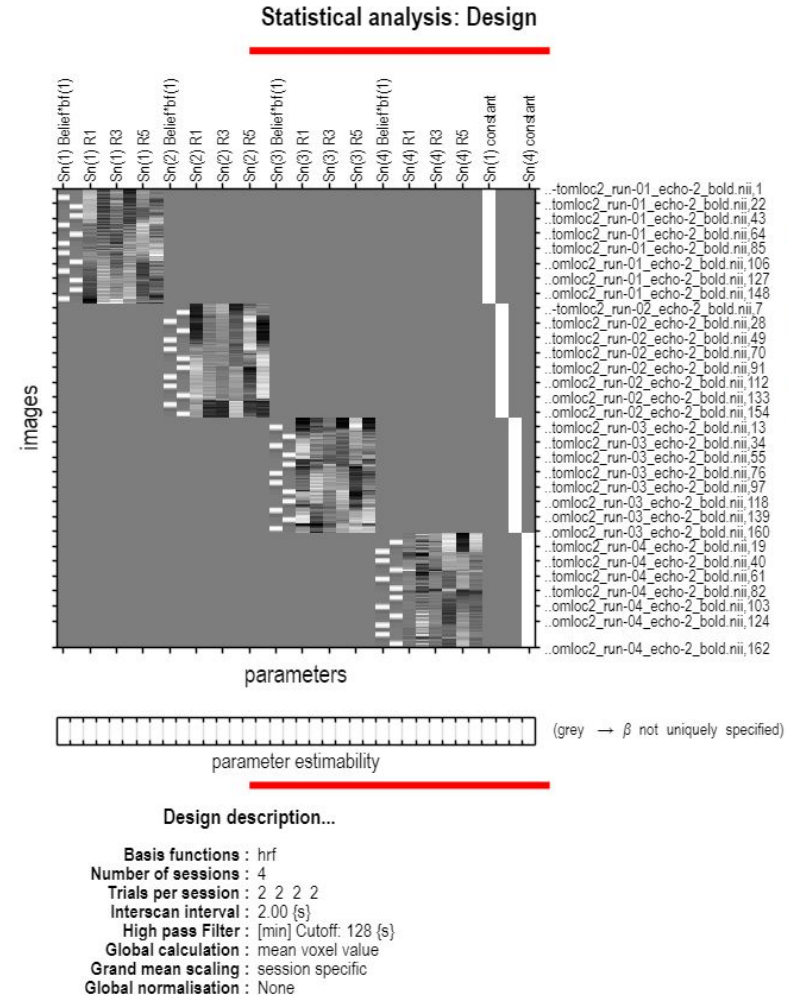
- Template_6 is in Windows
- all the u_rc files for subject 01~10 are in Windows
- all the ar_ files for subject 01~10 are in Linux

1st level specification

- units for design: seconds
- interscan interval: 2 (TR = 2)
- microtime resolution: 48 (48 slices)
- microtime onset: 1 (0 secs: 1, 17, 33 slice)

derivative of 1st level specification

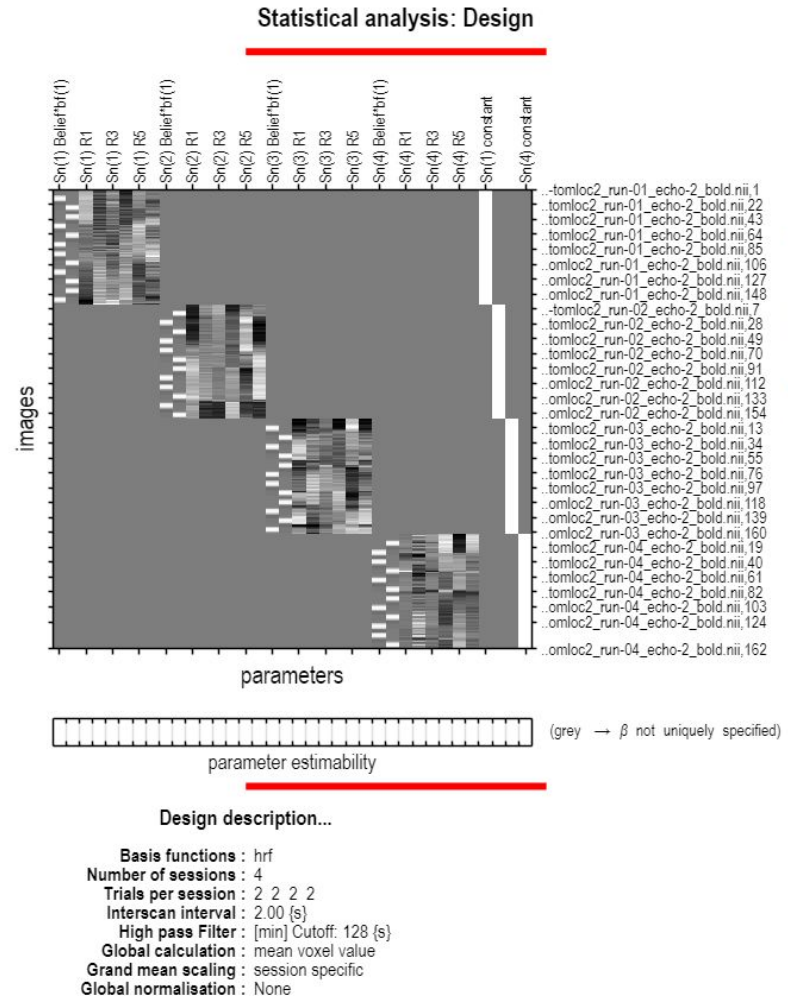
- an SPM.mat file for each subject



derivative of subject 01

4 runs, for each run:

- Belief
- Photo
- R1
- R2
- R3
- R4
- R5
- R6
- Constant



1st level estimation

- write residuals: no (I'm not doing functional connectivity analysis)
- method: classical (maximum likelihood estimation)

derivatives of 1st level estimation

- for each subject:
 - **beta**_0001.nii
 - **mask**.nii: 遮罩
 - **ResI**_0001.nii
 - **ResMS**.nii: residual
 - **RPV**.nii: multiple comparison

n coefficients: 1~36

n ? : 1~64

1st level contrast

- 2 contrasts: Belief, Photo
- Weight vector:
 - Belief: 1 0 0 0 0 0 1 0 0 0 0 0 1 0 0 0 0 0 1 0 0 0 0 0
 - Photo: 0 1 0 0 0 0 0 1 0 0 0 0 0 1 0 0 0 0 0 1 0 0 0 0

derivatives of 1st level contrast

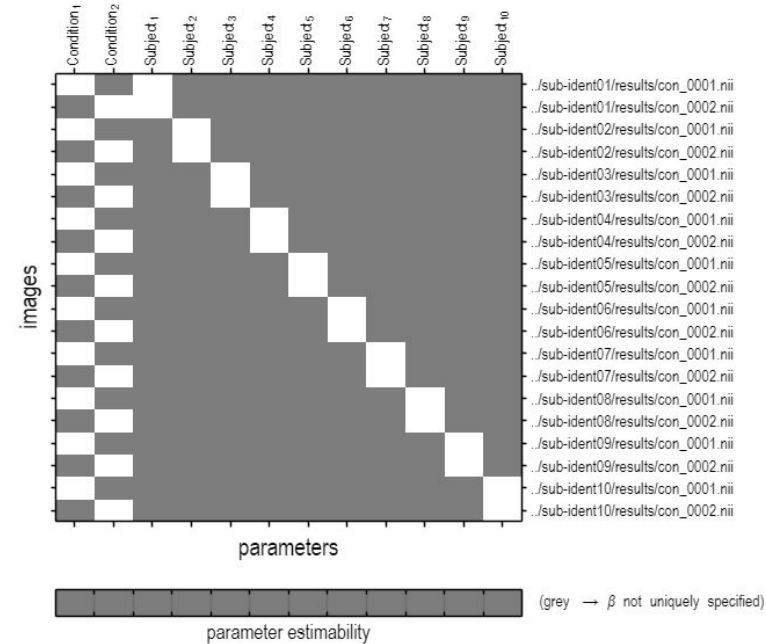
- for each subject:
 - con_0001.nii: Belief
 - con_0002.nii: Photo
 - spmT_0001.nii: Beliefi
 - spmT_0002.nii: Photo

2nd level specification

- paired-t test
 - 10 pairs
 - for each pair
 - scans[1,2]: sub-ident01/results/con_0001.nii, sub-ident01/results/con_0002.nii
 - grand mean scaling: no
 - ANCOVA: no
 - left other settings as default
- n subjects: 1~10

derivative of 2nd level specification

- SPM.file
- images: 20 contrasts nifti file
- parameters: 2 conditions, 10 subjects



Design description...

Design : Paired t-test
Global calculation : omit
Grand mean scaling : <no grand Mean scaling>
Global normalisation : <no global normalisation>
Parameters : 2 condition, +0 covariate, +10 block, +0 nuisance
 12 total, having 11 degrees of freedom
 leaving 9 degrees of freedom from 20 images

2nd level estimate

- SPM.file of 2nd level specification

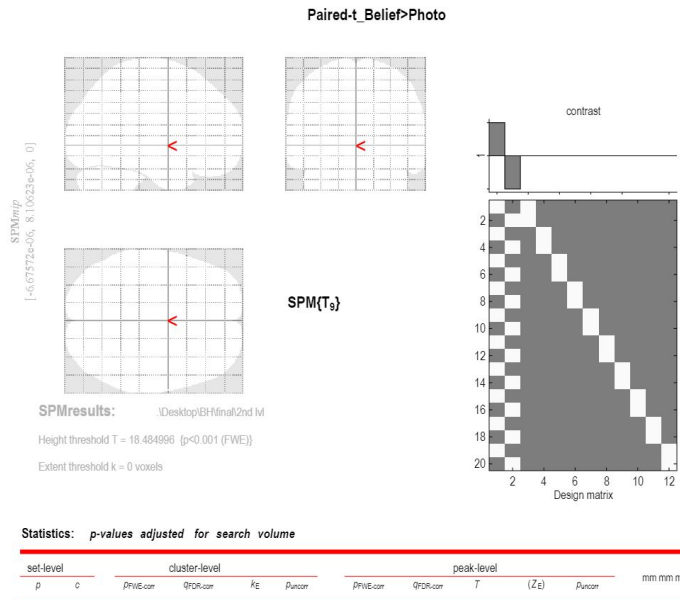
derivative of 2nd level estimate

n coefficients: 1~12

- beta_0001.nii
- mask
- ResMS
- RPV

Results

- SPM.file of 2nd level specification
- define new contrast
 - name: Paired-t_Belief>Photo
 - type: t-contrast
 - contrast: 1 -1
 - apply masking: none
 - p value: **.001**
- no suprathreshold clusters



no suprathreshold clusters

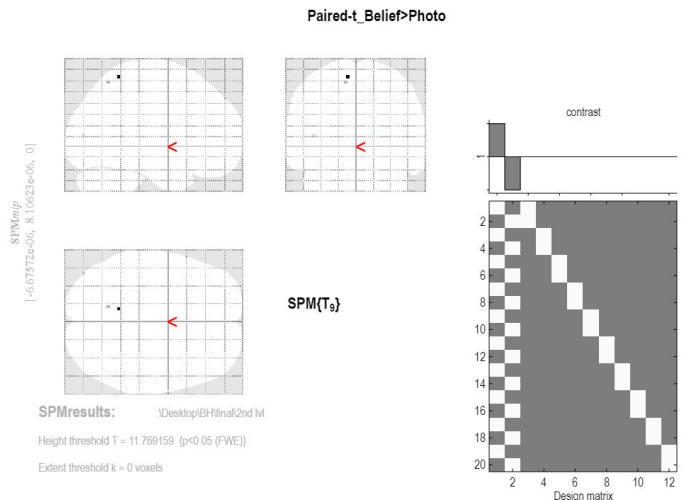
table shows 3 local maxima more than 8.0mm apart

Height threshold T = 18.48, p = 0.000 (0.001)
Extent threshold: k = 0 voxels
Expected voxels per cluster, <k> = 0.053
Expected number of clusters, <c> = 0.02
FWEp: 11.789, FDRp: Inf, FWEc: Inf, FDRc: Inf

Degrees of freedom = [1 0, 9 0]
FWHM = 11.2 11.1 10.3 mm mm mm; 4.7 4.6 4.1 (voxels)
Volume: 1585239 = 110086 voxels = 1115.0 resels
Voxel size: 2.4 2.4 2.5 mm mm mm; (resel = 89.28 voxels)

Results

- SPM file of 2nd level specification
- define new contrast
 - name: Paired-t_Belief>Photo
 - type: t-contrast
 - contrast: 1 -1
 - apply masking: none
 - p value: .05
- clusters
 - (-10, -50, 62)
 - (-12, -60, 58)
 - (-43, -84, 8)



Statistics: p-values adjusted for search volume

set-level		cluster-level				peak-level					mm mm mm
p	C	PFWE-cor	QFWE-cor	kE	Puncor	PFWE-cor	QFWE-cor	T	(Z)	Puncor	
0.002	3	0.007	0.031	1	0.031	0.011	0.845	14.05	5.20	0.000	-10 -50 62
		0.007	0.031	1	0.031	0.030	0.845	12.48	5.01	0.000	-12 -60 58
		0.007	0.031	1	0.031	0.039	0.845	12.13	4.96	0.000	-43 -84 8

table shows 3 local maxima more than 8.0mm apart

Height threshold: T = 11.77, p = 0.000 (0.050)
Extent threshold: k = 0 voxels
Expected voxels per cluster, <k> = 0.205
Expected number of clusters, <c> = 0.24
FWEp: 11.788, FDRp: Inf, FWEc: 1, FDRc: 1

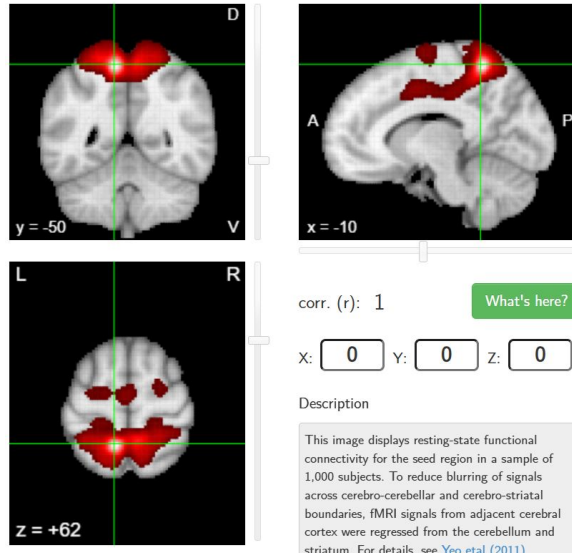
Degrees of freedom = [1 0, 9 0]
FWHM = 11.2 11.1 10.3 mm mm mm; 4.7 4.6 4.1 (voxels)
Volume: 1585239 = 110086 voxels = 1115.0 resels
Voxel size: 2.4 2.4 2.5 mm mm mm; (resel = 89.28 voxels)

$(-10, -50, 62)$

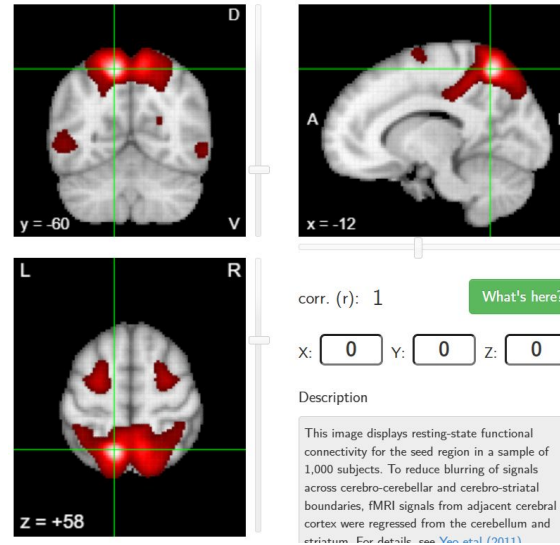
$(-12, -60, 58)$

occipital lobe
-> visual stimuli processing
 $(-43, -84, 8)$

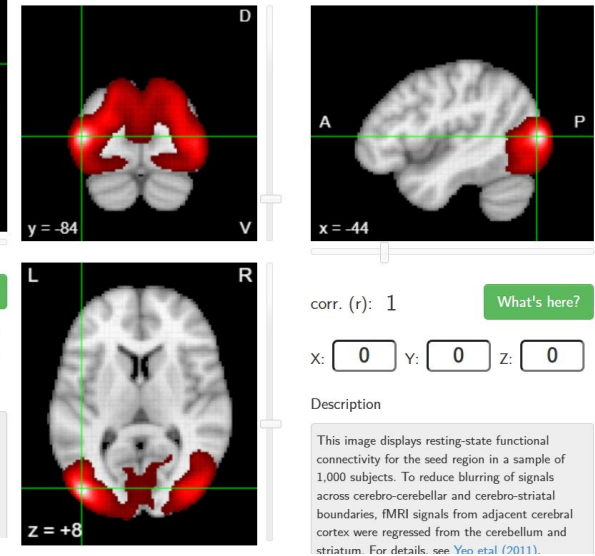
Functional connectivity and coactivation maps



Functional connectivity and coactivation maps



Functional connectivity and coactivation maps

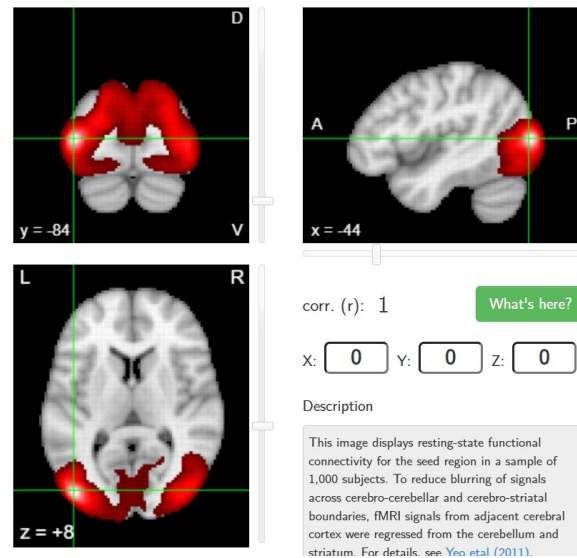


occipital lobe

- associative visual cortex
- visual stimuli processing
- the stimuli of Trial_{belief} may be more complex

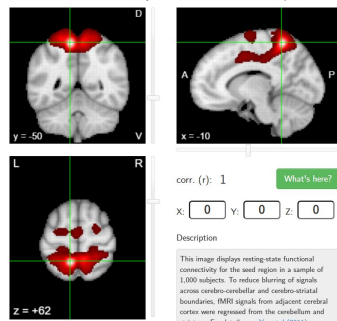
(-43, -84, 8)

Functional connectivity and coactivation maps



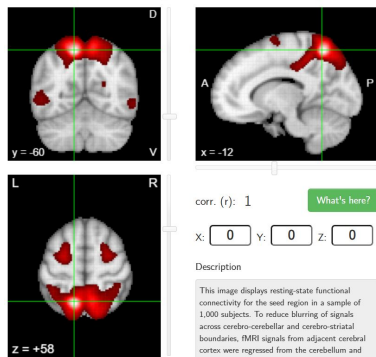
(-10, -50, 62)

Functional connectivity and coactivation maps



(-12, -60, 58)

Functional connectivity and coactivation maps



precuneus

- part of DMN
- central role in highly integrated tasks, including visuo-spatial imagery, episodic memory retrieval and self-processing operations, namely **first-person perspective taking** and an experience of agency.

Cavanna AE, Trimble MR. The precuneus: a review of its functional anatomy and behavioural correlates. *Brain*. 2006 Mar;129(Pt 3):564-83. doi: 10.1093/brain/awl004. Epub 2006 Jan 6. PMID: 16399806.

Original paper

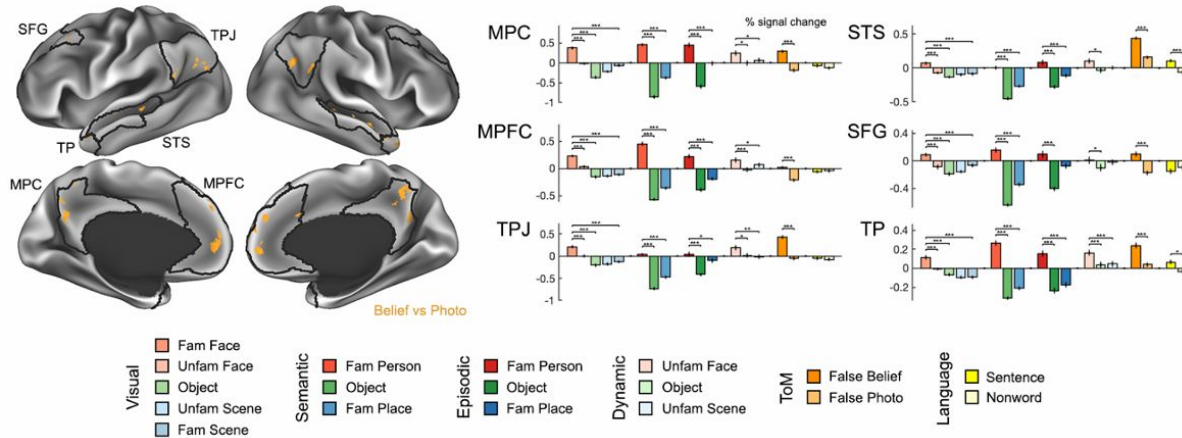


Figure S27: Theory of mind ROI analysis. Left: search spaces and functional ROIs from one representative participant. ROIs were defined as the top 5% of ToM-sensitive coordinates (false belief vs false photo contrast) within each search space. Right: Responses (% signal change) extracted from functionally defined ROIs. Error bars show standard error across runs. * $P < .05/7 = .0071$, ** $P < 10^{-3}$, *** $P < 10^{-4}$ (linear mixed model across runs, with participant included as random effect). Abbreviations: MPC, medial parietal cortex; MPFC, medial prefrontal cortex; SFG, superior frontal gyrus; STS, superior temporal sulcus; TPJ, temporo-parietal junction; TP, temporal pole.

Original paper

- Individual Functional ROI Analysis
- one subject
- apply mask

Mine

- whole brain analysis
- paired-t test for 10 subjects

next step

- ROI analysis